

Dummit & Foote 7.2

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1 Problems

Prop 4. Let R be an integral domain and let p and q be nonzero elements of $R[X]$. Then

- (1) $\deg(pq) = \deg(p) + \deg(q)$.
- (2) The units of $R[X]$ are just units of R .
- (3) $R[X]$ is an integral domain.

(1) The leading term of pq is the product of the leading terms of p, q . Since R has no zero divisors, the leading coefficient of the product is nonzero.

(2) Let $u, v \in R[X]$ be nonzero. Then $\deg(u) + \deg(v) \geq 0$ with equality iff $\deg(u) = \deg(v) = 0$, that is we only consider the inclusion of R in $R[X]$. The result follows.

(3) Let $f, g \in R[X]$ such that $fg = 0$. If either of f, g is nonzero and of degree at least 1, then the other must be zero. Otherwise we are restricted to R within $R[X]$, which is an integral domain.

Note trivial ring and zero ring are distinct. For any $n \geq 2$, if $M(R)$ is commutative, then R is trivial. Commutativity of $M(R)$ forces $ab = 0$ for all $a, b \in R$.

My example of a group ring (conjugation). Let G be a group. Define addition in R to be group multiplication, multiplication in R of $g, h \in G$ to be ghg^{-1} . Note $g(hk)g^{-1} = (ghg^{-1})(gkg^{-1})$. But multiplication is not associative. Group rings are of $\text{Hom}(A)$?

Definition (Group Ring): Let G be a finite group of order n and R be a commutative ring with identity. The group ring RG is the set of formal sums

$$a_1g_1 + a_2g_2 + \cdots + a_ng_n,$$

where addition is defined component-wise:

$$\sum_{i=1}^n a_i g_i + \sum_{i=1}^n b_i g_i = \sum_{i=1}^n (a_i + b_i) g_i,$$

and multiplication is defined by $(a_i g_i)(a_j g_j) = b g_k$, where $b = a_i a_j$ inside R and $g_k = g_i g_j$ in G , and then extending by distributive laws.

P 2. Let $p \in R[X]$ and $q \in R[X]$ be of minimal degree such that $pq = 0$. Suppose the leading terms of p, q are ax^n and bx^m . Then $ab = 0$. Consider the polynomial aq which is zero by minimality of q . We may repeat the argument $n - 1$ more times, for which we are left with $a_0 q = 0$. We have $a_i q = 0$ for all i . The first coefficient for each of these polynomials is $a_i b_m = 0$, so $b_m p = 0$.

On definition of ideal. Let R be a ring and $I \subset R$ be an additive subgroup of R . Let I be such that for all $a, b \in R$ and $i, j \in I$, then $(a + i)(b + j) - ab \in I$. We want $ib + aj + ij \in I$. Take $a, b = 0$, so I is closed under multiplication and is a subring of R . Take $i = 0$, so $aj \in I$ for all $a \in R$. I is a left ideal of R . Take $j = 0$, so $ib \in I$ for all $b \in R$, so I is a right ideal. If we suppose I is an ideal, then $ib, aj \in I$, so $ij + ib + aj \in I$. R/I has ring structure iff I is an ideal.

On the first isomorphism theorem. Suppose $\varphi : R \rightarrow S$ is a ring homomorphism. Then $R/\ker(\varphi) \cong \varphi(R)$. φ descends to an additive group homomorphism $\bar{\varphi} : R/I \rightarrow \varphi(R)$. Since I is an ideal, multiplication is preserved and $\bar{\varphi}$ is a ring homomorphism. By construction, $\bar{\varphi}$ is surjective. $\bar{\varphi}$ is also injective because $\varphi(a) = \varphi(b)$ iff a, b lie in the same I -coset, proving the result.

Second Isomorphism theorem. Let A be a subring and I be an ideal. Then $A + I$ is a subring and $A \cap I$ is an ideal of A and $(A + I)/I \cong A/(A \cap I)$.

$A + I$ is a ring because it is closed under addition. If $a + i, b + j \in A + I$,

$$(a + i)(b + j) = ab + ib + aj + ij \in A + I,$$

so $A + I$ is also closed under multiplication and thus a ring. Consider the projection $\pi : A \rightarrow (A + I)/I$. This is a well-defined map over R and therefore A . This map is surjective, so $(A + I)/I$ is a ring. $\ker(\pi) = A \cap I$, so $A \cap I$ is an ideal of A . By First Isomorphism theorem the result stands.

Third Isomorphism theorem. Consider the inclusion map $i : R/I \rightarrow R/J$. This map is surjective since R spans R/I . $\ker(i) = J/I$, so by First Isomorphism theorem $(R/I)/(J/I) \cong R/J$.

Fourth Isomorphism theorem. The map $A \leftrightarrow A/I$ is an inclusion-preserving map between the set of subrings A of R that contain I and the set of subrings of R/I . If $I \subset A \subset B \subset R$, I is an ideal of both A, B . We have $A/I, B/I$ are sensical rings, and $A/I \subset B/I$. Suppose $A/I \subset B/I$. For any $a \in A$ there exist $a + i = b + j$, or $a = b + (j - i)$. $I \subset B$, so $b + (j - i) \in B + B = B$. Hence, $A \subset B$.

P 16. Let $\varphi : R \rightarrow S$ be a ring homomorphism and $T = \varphi(Z(R))$, the image of the center of R . Then T is contained in $Z(S)$.

Let $s \in S$ and $t \in T$. Write $s = \varphi(r)$ and $t = \varphi(z)$ for some $z \in Z(R)$ and $r \in R$. Then

$$ts = \varphi(z)\varphi(r) = \varphi(zr) = \varphi(rz) = \varphi(r)\varphi(z) = st.$$

Hence, $t \in Z(S)$.

P 17. (a) Let R, S be nonzero rings with identity and denote their identities by 1_R and 1_S , respectively. Let $\varphi : R \rightarrow S$ be a nonzero ring homomorphism.

$$1_S\varphi(1_R) = \varphi(1_R1_R) = \varphi(1_R)\varphi(1_R),$$

so $(1_S - \varphi(1_R))\varphi(1_R) = 0_S$. $1_S - \varphi(1_R) \neq 0_S$ by assumption and $\varphi(1_R) \neq 0_S$, as otherwise φ would be the zero homomorphism. Hence, $\varphi(1_R)$ is a zero divisor of S . Integral domains do not have zero divisors, so $\varphi(1_R) = 1_S$.

(b) Let u be a unit of R and $v \in R$ such that $uv = vu = 1_R$. Then

$$\varphi(u)\varphi(v) = \varphi(v)\varphi(u) = 1_S.$$

Hence, $\varphi(u)$ is a unit of S . Also, $\varphi(u^{-1}) = \varphi(v)^{-1}$.

P 24: Let $\varphi : R \rightarrow S$ be a ring homomorphism.

(a) Prove that if J is an ideal of S , then $\varphi^{-1}(J)$ is an ideal of R . Apply to the special case $R < S$ and φ is the inclusion map to deduce if J is an ideal of S , $J \cap R$ is an ideal of R .

Let $a, b \in R$ such that $\varphi(a), \varphi(b) \in J$. Then $\varphi(a - b) = \varphi(a) - \varphi(b) \in J$. Note $\varphi(a\varphi^{-1}(J)) = \varphi(a)J = J$. Similarly, $\varphi(\varphi^{-1}(J)a) = J\varphi(a) = J$. If φ is the inclusion map, $\varphi^{-1}(J) = J \cap R$, which is to be an ideal of R by the previous work.

(b) $\varphi(a)\varphi(I) = \varphi(aI) = \varphi(I) = \varphi(I)\varphi(a)$. Take $R < S$ and φ to be the inclusion map. Pick $R \not\leq S$. In particular, take $R = L_i$ and $S = M_n(\mathbb{R})$. L_i is a one-sided ideal, but not two-sided.

P 27. Note $-1 = (-1)^2 = 1$, so $2 = 0$. $1 \neq 0$ since R is nonzero, so the characteristic of R is 2.

P 30. Let R be commutative. Let $f = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n \in R[X]$.

(a) Show that f is a unit iff a_0 is a unit and a_i is nilpotent for $i > 0$.

Suppose f is of the given form. a_0 is a unit and $f - a_0$ is nilpotent. The sum of a nilpotent and unit is a unit. f is a unit. Now sps f is a unit. Let g be such that $fg = 1$. $a_nb_m = 0$, $a_nb_{m-1} + a_{n-1}b_m = 0$, $a_nb_{m-2} + a_{n-1}b_{m-1} + a_{n-2}b_m = 0$. Eventually, $a_n^{m+1}b_0 = 0$. b_0 is a unit, so $a_n^{m+1} = 0$, and a_n is nilpotent. $f - a_nx^n$

is a unit and we repeat the process.

(b) Show that f is nilpotent iff all the coefficients are nilpotent.

Suppose f is nilpotent. There exists m such that $f^m = 0$. Consider the leading coefficient of f^m , which is zero. Hence, $a_n^m = 0$, so a_n is nilpotent. It follows $a_n x^n$ is nilpotent, so we consider the remaining n coefficients. By induction, all of the coefficients are nilpotent. On the contrary, if f has only nilpotent coefficients, let $m_0, m_1, m_2, \dots, m_n$ be such that $a_i^{m_i} = 0$. Let $M = m_0 + m_1 + \dots + m_n$. Consider f^M . Each coefficient is a binomial coefficient times some a_j to some power at least m_j by Pigeonhole, so $f^M = 0$.

P 34.

(a) Prove that $I + J$ is the smallest ideal containing I and J .

$i, j \in K$, so $i + j \in K$, or $I + J \subset K$. $(i + j) - (i' + j') = (i - i') + (j - j') \in I + J$, so $I + J \subset R$. $I + J \triangleleft R$ since $aI + aJ = I + J$.

(b) $x = i_1 j_1 + i_2 j_2 + \dots + i_n j_n \in I + I + \dots + I = I$. Similarly, $x \in J$, so $x \in I \cap J$. $aij \in IJ$. $ija \in IJ$.

(c) Let $I = 2\mathbb{Z}$, $J = 2\mathbb{Z}$, $R = \mathbb{Z}$, multiplication is natural. $IJ \subset 4\mathbb{Z}$, but $I \cap J = 2\mathbb{Z}$.

(d) Proofs assume the existence of 1. Consider $I = 4\mathbb{Z}$, $J = 6\mathbb{Z}$, $R = 2\mathbb{Z}$. $I \cap J = 12\mathbb{Z}$, $IJ \subset 24\mathbb{Z}$, $I + J = 2\mathbb{Z} = R$.